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The polarization mechanism in CdTe Schottky detectors

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Schottky CdTe nuclear detectors are affected by bias-induced polarization phenomena when operating at room temperature. A space charge buildup occurs at the blocking contact causing the degradation in detection performance. By means of Pockels effect, we study the electric field distribution inside the detector and its variation with time and temperature. The analysis of the space charge has allowed us to point out the role of the Schottky contact and of carrier detrapping from deep levels in the polarization mechanism. Moreover, measured current transients have been quantitatively accounted for by the increase in the electric field at the blocking junction. © 2009 American Institute of Physics. [DOI: 10.1063/1.3099051]

In x- and γ -ray semiconductor detectors the electric field distribution and its stability strongly affect the charge collection properties. In this respect, the material quality and the nature of the electrical contacts play a key role.

In the past decade, Schottky In/CdTe/Pt detectors with very promising spectroscopic performance have been developed.¹ This feature is mainly due to the use of indium as hole blocking contact. The high hole barrier height of this metal on CdTe, about 1 eV², allows very high average electric fields (several kV/cm) to be applied without a considerable current increase. Unfortunately, these In/CdTe/Pt detectors are not stable at room temperature and their charge collection properties degrade on the hour time scale showing a *polarization* effect.¹⁻³ In general, polarization, which is induced by applying a bias voltage to the detector¹⁻⁵ or irradiating it with high photon fluence,^{6,7} causes a progressive decrease in counting rate and pulse amplitude with time. In the first case, the effect is reversible, as bias resetting makes detection capabilities fully recovered, while in the second case the recovery after bias resetting is only partial.⁶ The voltage induced polarization has been observed through many years^{4,5} in CdTe detectors but its origin is still debated. There are indications^{2,4,5} that hole detrapping from deep acceptors by inducing a strong space charge buildup could be the responsible mechanism; on the other hand, the polarization effect seems to be very dependent on the electrode interfaces.⁸⁻¹⁰

In this work, we address the problem by studying the dynamics of the space charge distribution responsible for the voltage induced polarization. To this scope, we use the Pockels effect to precisely evaluate the electric field distribution inside an In/CdTe/Pt detector by the analysis of the near-infrared radiation (NIR) transmitted through the biased detector.

We report measurements which refer to a single AcroRad planar detector 10 × 10 mm² in size, 1 mm thick but similar results have been obtained on other nominally equal detectors. The material is a high quality (111)-oriented Cl doped semi-insulating (resistivity $\sim 10^9$ Ω cm, slightly *p*-type) CdTe single crystal grown by traveling heater method. The

anode contact is formed by depositing In on the Te-face of the crystal; a Pt layer, electroless deposited on the Cd-opposite face, acts as a cathode.

In our experiment, the NIR probe is linearly polarized at $\vartheta_1 = 45^\circ$ with respect to the (111) crystal direction along which the electric field is applied. The probe impinges on one detector surface (normal to the [111] electrode planes), and is then transmitted through a $\vartheta_2 = -45^\circ$ oriented analyzer. The transmitted probe intensity is digitized by a charge coupled device camera and stored as an image. For each (x, y) point of the detector side, the intensity $I(x, y)$ of the transmitted light depends on the mean electric field distribution $E(x, y)$ along the optical path, through the relation¹¹

$$I(x, y) = I_0(x, y) \sin^2 \left[\frac{\pi E(x, y)}{2 E_0} \right], \quad (1)$$

where $I_0(x, y)$ is the transmitted intensity with both the polarizers parallel, and E_0 is a constant dependent on material properties and on the wavelength of the probe radiation (980 nm, in our case). The electric field profiles $E(y)$ from anode to cathode are calculated by averaging $E(x, y)$ along the homogeneous x -direction, parallel to the electrodes. Preliminary measurements with different NIR intensities allowed us to verify that the electric field distribution is not perturbed by the radiation itself. By means of a Peltier cell-based system, measurements have been carried out at temperatures between 25 and 55 °C (stability ± 0.1 °C). More details on the setup have been previously published.¹²

The detector has been biased at 500 V and, by recording successive Pockels images,¹² the evolution of the internal electric field profiles has been calculated following Eq. (1). In general, the field is found to vary with time. In particular, in Fig. 1, which refers to room temperature, it is evident that just after the bias was applied, the field is greater at the anode, but it is not negligible at the cathode. With time, the field “moves” toward the positive In electrode, and in about 120 min it becomes confined in 2/3 of the detector thickness.

From the numerical derivative of the E -field profiles $dE/dx|_{x=0}$

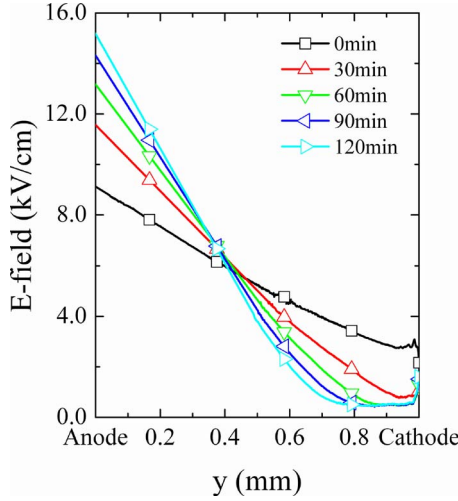


FIG. 1. (Color online) Electric field profiles from anode to cathode at different times. Detector is biased at 500 V. Room temperature (25 °C).

space charge increases with time evolving toward saturation values, and that the behavior is clearly thermally activated.

We remark that, according to the (slightly) *p*-type nature of CdTe material, the field decreases from the anode side, where the majority carriers are blocked by the In contact, and the net space charge is negative. Such charge can be mainly attributed to acceptors negatively ionized. Just after biasing few negative charges become immediately available due to the partial ionization of acceptors under equilibrium, but then most of the charge increases due to the hole detrapping ($N_A = N_A^- + h^+$). As a main consequence, the field becomes more and more confined near the anode. In agreement with the statistical nature of the detrapping process, the space charge follows an exponential behavior and the Poisson equation can hence be written at the anode as

$$\left. \frac{dE}{dx} \right|_{x=0} = -\frac{q}{\epsilon} \left[N_A^- \Big|_{x=0, t=0} + \Delta N_A^- \Big|_{x=0} (1 - e^{-t/\tau}) \right], \quad (2)$$

where τ is the hole detrapping time, $N_A^- \Big|_{x=0, t=0}$ and $\Delta N_A^- \Big|_{x=0}$ are the initial and incremental concentrations of ionized acceptors at the anode, respectively. Figure 2 shows that the best fit of these three parameters produces a good agreement with experimental data.

The time constants are well described by an Arrhenius plot, as shown in Fig. 3. From the exponential fitting of $(1/\tau T^2)$ versus $1/kT$, an activation energy of E_A

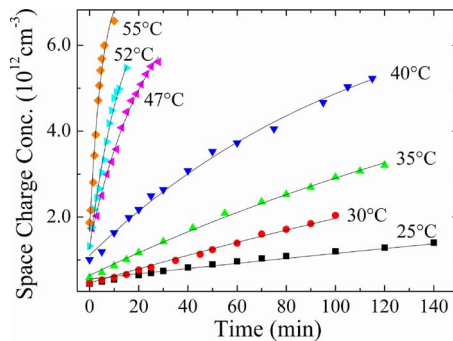


FIG. 2. (Color online) Time evolution of the space charge at the anode for the tested temperatures. Exponential fit to the data are also shown (solid lines).

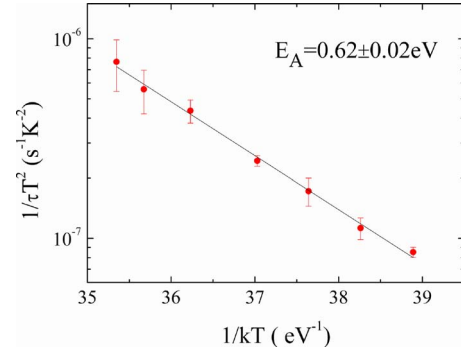


FIG. 3. (Color online) Arrhenius plot of the time constants.

$= 0.62 \pm 0.02$ eV is obtained. This energy is in good agreement with the value obtained by Toyama *et al.*² using a different approach based on the correlation between current transients and the polarization phenomena observed in similar CdTe detectors. This level could be ascribed to cadmium vacancies (V_{Cd}) that are known^{13,14} to be present as native defects in the crystals and to play a significant role in the electrical resistivity. It has been suggested that this level could be also responsible for polarization phenomena.¹⁴ This is reasonable because this level is close to mid gap and, hence, to the equilibrium Fermi level. When reverse bias is applied, free carrier depletion mainly affects the charge occupation of such mid gap levels through charge detrapping/space-charge buildup processes.

The thermal activation of the observed polarization process accounts for the improved stability of Schottky detectors at low temperature. Matsumoto *et al.*¹ observed stable performance for several days at -10 °C. This experimental finding is consistent with the value we extrapolate from our experimental time constants (4 days at -10 °C). Moreover, it has been often reported^{1,2,8} that longer detector stability is achieved when higher voltages are applied. This can be easily understood taking into account that the detector is irradiated from the cathode side. The higher the bias and the detector overdepletion, the later the electric field drops to negligible values at the cathode, which starts to degrade the detection performance.

In Fig. 2 it can be seen that the stationary concentrations of ionized acceptors are relatively low (between 10^{12} and 10^{13} cm⁻³) and they increase with temperature. This indicates that due to their deep energy level, acceptors do not become completely ionized, even at high temperature (55 °C). Full ionization is actually limited by the flowing current that tends to neutralize these deep acceptors. In particular, the effect of a high hole barrier is to decrease the current, and thus to enhance the acceptor ionization, with the consequent shrinking of the field toward the blocking contact. An evidence of this behavior can be found in standard Pt/CdTe/Pt detectors, whose stability at room temperature can be correlated with the high Pt work function (5.65 eV) compared to In (4.09 eV).

Finally, the increase with time of the electric field at the anode gives rise to current transients. This is due to the lowering of the Schottky barrier height with time as a consequence of the increase in the electric field at the anode. This effect has been previously observed^{2,8} in polarization phenomena and correlated with the evolution of the detection performance.² Here, we are able to directly correlate the cur-

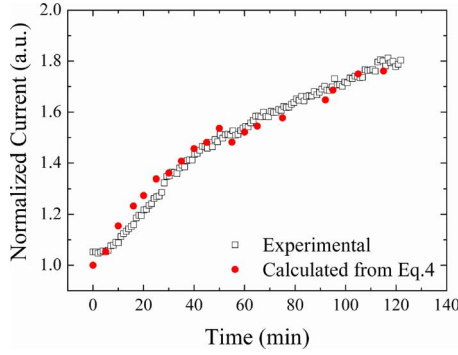


FIG. 4. (Color online) Current transients recorded during Pockels measurements along with the values calculated following Eq. (4). Data are normalized at the initial value. $T=40^\circ\text{C}$.

rent transients to the electric field E_{int} at the interface In/CdTe. It is well known that an interfacial layer is usually present at the metal/semiconductor interface, and that this results in the Schottky barrier to become field dependent. The barrier variation $\Delta\Phi$ is proportional to the field at the interface¹⁵

$$\Delta\Phi(t) = \alpha E_{\text{int}}(t), \quad (3)$$

where α is a constant related to the interfacial layer thickness. Since the reverse current depends exponentially on the total Schottky barrier height $\Phi = \Phi_0 - \Delta\Phi(t)$, the current transient can be expressed as

$$I(t) = I_0 e^{[q\alpha E_{\text{int}}(t)]/(kT)}, \quad (4)$$

where I_0 is a time-independent constant. Hence, a current increase with time is expected as a consequence of the field increase in the interface.

We compare in Fig. 4 the experimental current transients with those calculated inserting in Eq. (4) the $E_{\text{int}}(t)$ values obtained from the previous Pockels analyses. Plotted data refer to $T=40^\circ\text{C}$ with $\alpha=11.8\text{ nm}$. The comparison in Fig. 4 shows that from direct measurements of the electric field at the anode, the current transient is well described by Eqs. (3) and (4) with no further assumption. Similar results have been obtained at different temperatures, with a slightly scattered distribution of the best fitting α value, namely, $\alpha = 10.5 \pm 2.1\text{ nm}$.

It is worth to note that two competing effects are at work: the hole detrapping enhances the acceptor ionization leading to an increase in the electric field at the anode; how-

ever, such increase causes a current increase which in turn leads to an acceptor ionization decrease. Hence, the system attains stationary conditions at a specific acceptor ionization level. A further evidence of the correlation between polarization and current transients comes from the very small current transients observed in the nonpolarizing Pt/CdTe/Pt detectors (not shown here).

In conclusion, exploiting the Pockel effect, we measured the electric field profiles as a function of time and temperature. This allows the monitoring of the space charge dynamics at the anode and the identifying of the mechanism leading to the observed polarization effect. The electric field distribution is found to be governed by the ionized deep acceptors, which are bulk defects probably associated to native cadmium vacancies. The interface at the anode plays a fundamental role in controlling the flowing current and therefore the extent of hole detrapping which is, ultimately, the cause of the polarization effect in CdTe detectors.

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